



**PrintLog3D.com**  
MATERIALS. SETTINGS. LOGS. RESULTS.

FREE PRINTABLE FIELD GUIDE

# FILAMENT SETTINGS FIELD GUIDE

17 MATERIALS. ONE REFERENCE. LESS GUESSWORK.

Start with the published range, record what actually works on your machine, and build a repeatable settings history instead of printing from memory.



## USE THE GUIDE LIKE A DATA LOOP

01

### CHOOSE

Find your material and start inside the published range.

02

### PREP

Meet the enclosure, cooling, and drying notes before tuning.

03

### PRINT

Run a controlled test on your actual printer and filament.

04

### LOG

Write down the settings and the result - good or bad.

05

### TUNE

Change deliberately, then keep the setting that earned the result.

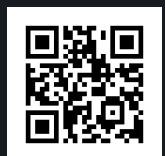
#### STARTING POINTS, NOT MAGIC NUMBERS

These are typical manufacturer-published ranges - not measurements taken by PrintLog3D. Always check your filament maker's own figures.

#### LIVE MATERIAL LIBRARY

Guides, material pages, updates

[PRINTLOG3D.COM](https://printlog3d.com)



PRACTICAL SETTINGS. REAL RESULTS. NO MORE PRINTING FROM MEMORY.



# 17-MATERIAL SETTINGS MATRIX

Typical published ranges at a glance. Use the material maker as the final authority.

NO ENCLOSURE

ENCLOSURE REQUIRED

DRYING GUIDANCE

| MATERIAL   | NOZZLE C | BED C   | ENCLOSURE   | PART COOLING | DRY AT                 |
|------------|----------|---------|-------------|--------------|------------------------|
| PLA        | 180-220  | 50-60   | Not needed  | Yes (100%)   | 45C / 4-6h (opt.)      |
| PETG       | 220-250  | 70-85   | Not needed  | Yes (30-50%) | 65C / 4-6h             |
| ABS        | 220-250  | 100-110 | REQUIRED    | No / low     | 65-80C / 2-4h          |
| ASA        | 235-255  | 90-110  | REQUIRED    | No / low     | 70-80C / 4h            |
| PC         | 260-310  | 100-120 | REQUIRED    | No           | 80C / 6-8h             |
| PEEK       | 370-420  | 120-160 | REQUIRED    | No           | 120-150C / at least 3h |
| Nylon PA6  | 230-260  | 70-90   | REQUIRED    | Minimal      | 70-80C / 8-12h         |
| Nylon PA12 | 240-270  | 70-90   | REQUIRED    | Minimal      | 70-80C / 8-12h         |
| PA-CF      | 260-280  | 80-100  | REQUIRED    | Minimal      | 70-80C / 8-12h         |
| PETG-CF    | 240-260  | 80-90   | Not needed  | Yes (30-50%) | 65C / 4-6h             |
| PCTG       | 230-250  | 70-85   | Not needed  | Yes (30-50%) | 65C / 4-6h             |
| CPE        | 240-260  | 75-90   | RECOMMENDED | Yes (low)    | 65-70C / 4-6h          |
| HIPS       | 220-250  | 100-115 | REQUIRED    | No / low     | 65C / 4h (opt.)        |
| PLA Silk   | 195-225  | 50-60   | Not needed  | Yes (50-70%) | 45C / 4-6h (opt.)      |
| PLA Matte  | 195-220  | 50-60   | Not needed  | Yes (100%)   | 45C / 4-6h (opt.)      |
| PLA Wood   | 190-220  | 50-60   | Not needed  | Yes          | 45C / 4-6h (opt.)      |
| PLA Metal  | 190-225  | 50-60   | Not needed  | Yes          | 45C / 4-6h (opt.)      |

**IMPORTANT: RANGE DOES NOT MEAN "BEST SETTING"**

Printer geometry, filament brand, moisture, part design, and environment can change the useful setting. Use this matrix to choose a safe starting range, then log the result on your own machine.



# FAILURE MODES: THE FIRST THING TO CHECK

The most common problem called out for each material in the source guide - tightened into field-note format.

**PLA****HEAT SOFTENING**

Parts soften around 60C. If the part must survive heat, move to PETG or ABS instead of trying to tune PLA.

**PETG-CF****WRONG EXPECTATION**

Carbon fiber makes the material stiffer, not stronger in every sense. Impact resistance goes down, not up.

**PETG****STRINGING**

Lower nozzle temperature 5C at a time and raise travel speed before changing retraction distance.

**PCTG****AVAILABILITY**

The problem may be sourcing more than printing. Fewer brands carry it, so color choice is narrower than PETG.

**ABS****WARPING / LAYER SPLIT**

Corners lift and layers split when the print is exposed to drafts. Enclose the printer and turn part cooling off.

**CPE****NAME CONFUSION**

In 3D printing, CPE means co-polyester. Do not confuse it with chlorinated polyethylene, an unrelated industrial rubber.

**ASA****WARPING + ODOR**

Similar warping behavior to ABS, with a stronger smell. Ventilate the room and keep the chamber warm and still.

**HIPS****SUPPORT ASSUMPTIONS**

Do not buy it as support material unless you have a second extruder. On a single-nozzle printer it is simply an ABS-like filament.

**PC****LAYER DELAMINATION**

A clean-looking print can still delaminate under load when the filament is wet. Dry it and keep it dry during printing.

**PLA Silk****LOST SHEEN**

Too much cooling dulls the sheen. Reduce fan, print a little hotter, and slow down versus plain PLA.

**PEEK****MACHINE LIMITS**

The main problem is often machine capability. If your hot end or bed cannot reach the required temperatures, tuning cannot solve it.

**PLA Matte****ABRASIVE FILLER**

The matting filler is mildly abrasive. Brass can work, but expect faster nozzle wear than plain PLA.

**Nylon PA6****MOISTURE**

Steam pops and a furry surface mean the filament is wet. Dry once is not enough - it needs to stay dry during the print.

**PLA Wood****CLOGGING**

Wood particles bridge small nozzles easily. Use 0.4mm or larger and avoid long retractions.

**Nylon PA12****BED ADHESION**

Nylon does not like PEI. A garolite or PA-specific sheet, or a glue-stick layer, can solve what temperature tuning will not.

**PLA Metal****STRENGTH MYTH**

Metal filler adds weight and finish, but reduces strength compared with plain PLA.

**PA-CF****NOZZLE WEAR**

Underextrusion that worsens over time can mean the carbon fiber has widened the nozzle bore. Use hardened steel or ruby before the first print.

**THE GOAL IS NOT TO MEMORIZE FIXES. BUILD A REPEATABLE HISTORY.**

Record the setting, the failure, the change, and the result on the print log.

[PRINTLOG3D.COM](https://printlog3d.com)



# LOG IT. LEARN IT. PRINT BETTER.

A clean record turns one good print into a repeatable process - and one failed print into useful evidence.

## PRINT LOG - ONE RUN

MATERIAL + SETTINGS + RESULT

### MATERIAL

e.g. PETG

### BRAND / SPOOL

brand, color, batch

### PRINTER

machine / profile

### NOZZLE

diameter + material

### NOZZLE

\_\_\_\_\_ C

### BED

\_\_\_\_\_ C

### FAN

\_\_\_\_\_ %

### SPEED

\_\_\_\_\_ mm/s

### PREP

☐ ENCLOSURE

☐ DRIED

☐ COOLING

### RESULT

☒ CLEAN

☐ USABLE

☐ FAIL

### WHAT HAPPENED?

### NEXT CHANGE - ONE THING ONLY

## FIELD REMINDERS

Pulled directly from the material notes.

**PLA** Heat sensitive near 60C.

**ABS / ASA** Control drafts; enclosure matters.

**NYLON** Keep it dry while printing.

**PA-CF** Use a hardened or ruby nozzle.

**PEEK** Machine capability is the gate.

**PLA WOOD** Use 0.4mm or larger nozzle.

**PETG-CF** Stiffer does not mean tougher.

## KEEP THE LIVE REFERENCE HANDY



Scan for the material library  
[PRINTLOG3D.COM](https://printlog3d.com)

## 17 MATERIALS ONE SETTINGS REFERENCE



Free to browse. Built to log.

**PRACTICAL SETTINGS. REAL RESULTS.**